Healthcare IT Fundamentals - Lecture 3

Healthcare Terminology, Standards & Data Exchange (Simple Notes)

Big Picture

In the previous lectures, we learned:

1. Why healthcare moved from paper to digital
2. HIPAA, HITECH, compliance, and security

Now we learn:

✅ What EMR, EHR, and PHR mean

✅ How healthcare data is coded

✅ How systems exchange data

✅ How access is controlled

These are the building blocks of modern Healthcare IT.

---

1. EMR vs EHR vs PHR

This is one of the most commonly asked interview questions.

EMR (Electronic Medical Record)

Think:

> Digital version of a doctor's paper chart

Contains records from only one clinic or provider.

Example

You visit Dr. Smith's clinic.

The clinic stores:

- Diagnoses
- Prescriptions
- Visit history

inside its own system.

That is an EMR.

Limitation

Usually stays within one organization.

Not designed for easy sharing.

EHR (Electronic Health Record)

Think:

> Complete health history that follows the patient.

An EHR collects information from:

- Hospitals
- Clinics
- Labs
- Pharmacies
- Specialists

Example

Patient visits:

1. Family doctor
2. Cardiologist
3. Lab
4. Pharmacy

All records are combined into one view.

That's an EHR.

Key Feature

Interoperability

Can share information between organizations.

PHR (Personal Health Record)

Think:

> Health record managed by the patient.

Patient controls the data.

Example

Apps like:

- Apple Health
- Google Health (historically)
- Fitness apps
- Wearables

can contribute to a PHR.

Data includes:

- Steps walked
- Heart rate
- Blood pressure
- Personal notes

Quick Comparison

Feature	EMR	EHR	PHR
Maintained by	Doctor/Clinic	Multiple Providers	Patient
Shared Across Organizations	No	Yes	Patient Controlled
Purpose	Local Treatment	Complete Health View	Self Management

---

2. Medical Coding Standards

Healthcare uses standard codes.

Why?

Because everyone must understand the same diagnosis and procedure.

ICD-10

International Classification of Diseases

Used for:

✅ Diagnoses

Answers:

> What disease does the patient have?

Example

Patient has diabetes.

ICD-10 assigns a standard code.

Insurance companies, hospitals, and researchers all understand it.

CPT

Current Procedural Terminology

Used for:

✅ Procedures

Answers:

> What treatment was performed?

Examples

- Surgery
- Consultation
- X-Ray
- Blood test

Easy Memory

ICD = Disease

CPT = Procedure

SNOMED CT

Used for:

Detailed clinical documentation.

Provides richer clinical descriptions.

Example

Instead of simply recording:

```
Asthma
```

SNOMED can capture:

```
Severe persistent asthma
with acute exacerbation
```

More detail.

More accuracy.

LOINC

Logical Observation Identifiers Names and Codes

Used for:

Laboratory tests and observations.

Example

Blood glucose test.

Hospital and laboratory both understand the exact same test.

LOINC ensures consistency.

Coding Summary

Standard	Purpose
ICD-10	Diseases
CPT	Procedures
SNOMED CT	Clinical Terms
LOINC	Lab Tests

---

3. Why Coding Matters

Coding affects:

Billing

Insurance reimbursement depends on codes.

Research

Researchers analyze disease trends using codes.

Public Health

Governments track outbreaks using coded data.

Example:

Thousands of flu diagnoses with same ICD code.

Health officials detect an outbreak quickly.

---

4. Data Transport Standards

Having data is useless if systems cannot exchange it.

Transport standards allow systems to communicate.

Think:

> APIs for healthcare.

HL7

Health Level Seven

Traditional healthcare messaging standard.

Used for decades.

Examples

Hospital sends:

- Admission message
- Discharge message
- Lab result

to another system.

HL7 handles that communication.

Analogy

Like XML-based integration in enterprise software.

FHIR

Pronounced:

FIRE

Fast Healthcare Interoperability Resources

Modern healthcare standard.

Why Important?

Uses modern web technologies.

Similar to:

- REST APIs
- JSON
- Mobile apps

Example

Patient opens a mobile healthcare app.

The app retrieves records through FHIR APIs.

Software Engineer Analogy

Traditional HL7:

```
Legacy SOAP/XML
```

FHIR:

```
Modern REST API + JSON
```

Why Healthcare Loves FHIR

- Easier integration
- Mobile friendly
- Faster development
- Better interoperability

---

5. DICOM

Digital Imaging and Communications in Medicine

Used for medical imaging.

Examples

- X-rays
- CT scans
- MRI scans
- Ultrasound images

Purpose

Ensures images can be viewed anywhere.

Regardless of:

- Vendor
- Hospital
- Imaging system

---

6. Common Healthcare Acronyms

Healthcare loves acronyms.

CPOE

Computerized Provider Order Entry

Doctors enter orders electronically.

Examples:

- Medication orders
- Lab requests
- Radiology requests

Before

Handwritten prescription.

Problems:

- Hard to read
- Errors

After

Electronic order.

Safer and faster.

HIE

Health Information Exchange

Sharing patient data across organizations.

Example

Hospital sends patient records to another hospital.

That sharing network is an HIE.

CDS

Clinical Decision Support

Smart alerts that help clinicians.

Examples

System warns:

```
Drug A and Drug B
should not be taken together
```

or

```
Patient is allergic
to this medication
```

Think

Healthcare version of:

```
lint warnings
```

for medical decisions.

RCM

Revenue Cycle Management

The financial workflow.

From:

Patient Registration

↓

Treatment

↓

Billing

↓

Insurance Payment

↓

Final Collection

---

7. DIKW Pyramid

One of the most important concepts.

D = Data

Raw facts.

Example:

```
102°F
```

Just a number.

I = Information

Add context.

```
Patient's temperature is 102°F
```

Now it means something.

K = Knowledge

Analyze multiple pieces of information.

Example:

Many patients have:

- Fever
- Cough
- Fatigue

Knowledge:

```
Possible flu outbreak
```

W = Wisdom

Take action.

Example:

```
Start isolation procedures
```

Easy Example

Level	Example
Data	102
Information	Temperature = 102°F
Knowledge	Patient likely has fever
Wisdom	Start treatment

8. RBAC

Role-Based Access Control

One of the most important security models.

Problem

Not everyone should access everything.

Solution

Access is based on role.

Example

Nurse

Can access:

- Clinical notes
- Medication records

Doctor

Can access:

- Full patient record

Billing Staff

Can access:

- Insurance data
- Billing information

Cannot access:

- Detailed clinical notes

IT Admin

May manage systems

But cannot automatically view all patient records.

Software Engineer Analogy

```
role = "doctor"
```

Doctor permissions:

```
view_patient_record
update_diagnosis
prescribe_medication
```

```
role = "billing"
```

Billing permissions:

```
view_insurance
create_invoice
```

No access to treatment notes.

Why RBAC Matters

Supports HIPAA's:

Minimum Necessary Rule

People should only access data needed for their job.

Final Revision Sheet

EMR

Electronic record within one provider.

EHR

Shared health record across providers.

PHR

Patient-managed health record.

ICD-10

Disease codes.

CPT

Procedure codes.

SNOMED CT

Clinical terminology.

LOINC

Lab test coding.

HL7

Traditional healthcare messaging standard.

FHIR

Modern API-based interoperability standard.

DICOM

Medical imaging standard.

CPOE

Electronic ordering system.

HIE

Health Information Exchange.

CDS

Clinical Decision Support.

RCM

Revenue Cycle Management.

DIKW

Data → Information → Knowledge → Wisdom.

RBAC

Role-Based Access Control.

For a Ruby on Rails / Python Developer

Think of healthcare systems like this:

Software World	Healthcare World
Local Database	EMR
Distributed System	EHR
User Profile App	PHR
REST API	FHIR
Legacy Integration	HL7
Image Format	DICOM
RBAC	HIPAA Access Control
Validation Rules	Clinical Decision Support
Analytics Dashboard	DIKW Model

One-line takeaway:

EMR stores data, EHR shares data, FHIR moves data, ICD/CPT describe data, and RBAC protects data. 🚀